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New trends in autologous and allogeneic stem cell transplantation—an update from the annual ASH meeting 2020

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Summary At the 62nd annual meeting of the American Society of Hematology (ASH) 2020—which took place virtually for the first time—numerous clinical studies and research results were presented. The latest trends and exciting developments in the field of stem cell transplantation can be found in the following article, e. g. how transplantations with a mismatch donor can be safely performed, important studies regarding MDS and ALL patients, a promising second-line treatment in chronic GVHD, why the nutritional status and the microbiome should be part of the pre-transplant work-up and that lifelong follow-up for children and young adolescents after an allogeneic transplantation is essential.

Keywords Universal donor · High-risk MDS · Ruxolitinib · Multiple myeloma · Microbiome

Universal donor

Several oral presentations and one education session discussed whether it would be possible to increase the availability of suitable donors for allogeneic stem cell transplantation (SCT), especially for ethnic minorities who are often not sufficiently represented in national registries. In a prospective phase II study [1] of the Center for International Blood and Marrow Transplant Research (CIBMTR) from 2016–2019, only mismatch donors with 1-, 2-, 3- or 4-antigen mismatches, who are usually not first choice or are not used at all, were examined in both the myeloablative ($n=40$) and also in the reduced intensity (RIC) setting ($n=40$). Bone marrow (BM) was the preferred stem cell source. Immunosuppression consisted of a combination of sirolimus/mycophenolate mofetil (MMF) and posttransplant cyclophosphamide (PT-Cy). In a multivariate analysis, results were compared with a recent control group of different donor groups (mismatch donors with peripheral blood stem cells (PBSC) and standard immunosuppression, haploidentical donors with BM or PBSC and PT-Cy). There was no difference in 1-year overall survival (OS), progression-free survival (PFS), nonrelapse mortality (NRM) or grade II–IV acute graft vs. host disease (GvHD).

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Allogeneic stem cell transplantation in myelodysplastic syndrome

Early SCT is often not discussed with older patients with myelodysplastic syndrome (MDS) in clinical routine because the advantage over conventional therapies could only be shown in small and/or retrospective studies. A multicenter US study investigated the benefits of allogeneic SCT in 50- to 75-year-old MDS patients [2].

A total of 384 patients with high-risk (IPSS Int-2/high-risk) MDS were included. An HLA-identical donor was found for 260 patients within 90 days and then a RIC-SCT was performed, the remaining 124 patients received standard therapy with hypomethylating agents or only best supportive care (BSC), depending on the institutional guidelines. Survival analyses showed a clear benefit in favour of SCT, the 3-year OS in the transplant arm was 47.9% vs. 26.6% in the control arm (in all subgroups, including patients >65 years of age). No difference could be found in the quality of life, using standardized questionnaires.

Autologous vs allogeneic transplantation in acute lymphoblastic leukaemia

Numerous studies by the EBMT (European Society for Blood and Marrow Transplantation) were also presented at this year's ASH meeting. Sebastian Giebel showed a comparison between autologous (ASCT) and allogeneic SCT in acute lymphoblastic leukaemia (ALL) in first complete remission (CR). There is already promising data for ASCT in younger ALL patients; here he analysed patients with a median age of 60 years between 2000 and 2018. In all, 418 patients with RIC-SCT (donors all HLA-identical) and 142 patients with ASCT were examined retrospectively [3]. The 5-year OS showed no difference with 45% for ASCT vs. 42% for allogeneic SCT ($p=0.23$). It should be emphasized that the conditioning regimen in ASCT mainly included total body irradiation (in combination with chemotherapy) and 79% of the patients were minimal residual disease (MRD) negative. These two aspects seem to be important for successful ASCT in first CR in ALL patients >55 years.

Ruxolitinib in chronic graft vs. host disease

Ruxolitinib is a very potent treatment for acute GvHD, the immunosuppressive potential has already been shown in the REACH1 + 2 study. The REACH3 study (phase III) investigated ruxolitinib in chronic GvHD [4]. First-line treatment still comprises steroids, but 50% of the patients are refractory or dependent and so far there are no standardized recommendations for second-line treatment. The study was designed for these patients (steroid-refractory or dependent chronic GvHD) and, after randomization, either ruxolitinib at a dose of 10 mg twice a day (for 165 patients), in addition to the existing immunosuppression, or another second-line treatment (total of 10 options, for 164 patients) was initiated. Ruxolitinib was administered for 6 cycles of 28 days each; the primary endpoint of the study was the overall response rate (ORR; complete and partial remissions) according to NIH consensus criteria at the end of the last cycle (= week 24 after the start of therapy). The ORR with ruxolitinib of 49.7% was significantly higher than 25.6% in the control arm, with an expected side effect profile (including haematotoxicity, increased liver transaminases). Regarding infections, no difference was observed in the clinical course. REACH3 is the first phase III study to show the superiority of second-line treatment with ruxolitinib in chronic GvHD.

Autologous transplantation in multiple myeloma

In recent years, benefit and optimal timing of ASCT in multiple myeloma (MM) as part of the first-line treatment has been questioned in several publications. Therefore, long-term results of the well-known prospective, randomized IFM 2009 study [5] were presented at ASH 2020 (8 cycles VRD vs 3 cycles VRD + ASCT + 2 cycles VRD consolidation; 12 months of lenalidomide maintenance for both groups). The first interim analysis was published in 2017 in the *New England Journal of Medicine* and showed a significantly longer PFS in favour of transplantation (50 vs. 36 months). The study has now a follow-up of 9 years. PFS remains significantly longer (47.3 vs. 35 months), while 8-year OS was not different (62 vs. 60%). MRD was an independent factor for prognosis and survival. Interestingly, patients in the ASCT arm achieve even deeper remissions because MRD-negative patients had a better survival here than MRD-negative patients in the VRD arm.

Autologous transplantation in aggressive B-cell lymphoma

Based on preliminary work, an Italian working group showed a poster presentation where 63 patients >60 years of age with aggressive B-cell lymphoma underwent ASCT as a consolidation therapy [6]. Sixty-five percent of the patients had a high-intermediate or high-risk international prognostic index (IPI). Immunohistochemically, 81% had a germinal-center B-cell (GCB)-like subtype; genetically 34 patients had a double hit, 10 patients a triple hit and 19 patients a c-MYC-ICN (ICN = increased copy number) B-cell lymphoma. First-line therapy consisted of R-DA-EPOCH. The ORR was 81% (68% CR rate), the 3-year PFS and OS were 67 and 69%, respectively. Ten patients were chemorefractory and ultimately all died of lymphoma. A total of 24 patients (17 in CR, 6 in PR, 1 with PD) underwent ASCT, which was significantly associated with better survival. The 3-year PFS and OS were 100 and 76% for patients in CR.

Impact of nutritional status and loss of microbiota diversity on survival after stem cell transplantation

In addition to known factors such as conditioning intensity and the influence of donor type (haploidentical vs. sibling vs. unrelated donors) on survival after SCT, supportive factos such as nutritional status and intestinal microbiome are increasingly important. Numerous works on the prognostic significance have already been shown, in particular by Marcel van den Brink from the MSKCC. He was able to demonstrate that a loss of intestinal microbiome diversity is associated with reduced survival due to an increased TRM. This not only applies to allogeneic, but also to ASCT.

A German study [7], which was presented as a poster, examined the influence of the nutritional status of AML patients who underwent allogeneic SCT, a total of 662 patients (median age 59 years) in CR (68%), CRi (14%), or with active disease (18%). According to ELN criteria, 25% were good, 30% intermediate, and 45% high risk. Obese patients (at diagnosis) had a higher NRM and shorter OS after SCT, and weight loss (ΔBMI > –2) between diagnosis and SCT was an independent risk factor for increased NRM and shorter OS.

Long-term follow-up

As part of the BMT Survivor Study (BMTSS), 580 allogeneic transplanted young adults ≤22 years of age who had survived at least 2 years after SCT were asked about "chronic health impairments" (CHC), e.g. diabetes mellitus, hypertension, thyroid diseases [8]. Results were then compared with those of their healthy, untreated siblings. Median age at time of SCT was 11.5 years, median follow-up was 10.7 years and the transplantations were between 1974 and 2014. Diagnoses were ALL (29%), AML/MDS (28%), SAA (13%), and others (30%). Incidence of grade 3–4 CHC in the 4th decade of life in survivors after SCT was significantly higher than in the healthy, untreated siblings (49.1 vs. 6.9%). Especially after total body irradiation the risk was significantly increased. This work underscores the need for lifelong follow-up in specialized institutions.

Take home messages

- Transplantation with mismatch donors combined with PT-Cy as immunosuppression can be safely performed.
- Elderly, fit patients with high-risk MDS and a suitable donor should not be withheld from an allogeneic SCT.
- In ALL patients in 1st CR with MRD negativity, a retrospective analysis showed that ASCT can lead to results comparable to those of allogeneic transplantation—a prospective randomised trial should be performed.
- Ruxolitinib is a promising second-line treatment in steroid-refractory or -dependent patients with chronic GVHD.
- ASCT in first-line treatment in MM was confirmed at ASH 2020.
- Nutritional status and the microbiome should be part of the pretransplant work-up.
- Lifelong follow-up for children and young adolescents after an allogeneic SCT is essential.

Conflict of interest A. Böhm and F. Keil declare that they have no competing interests.

References

1. Shaw BE, Jimenez-Jimenez A, Burns LJ, et al. Bridging the gap in access to transplant for Underserved minority patients using mismatched unrelated donors and post-transplant cyclophosphamide: a national marrow donor program/be the match (NMDP/BTM) initiative. 62nd ASH Annual Meeting, December 5–8, 2020; Abstract No. 297.
2. Nakamura R, Saber W, Martens MJ, et al. A multi-center biologic assignment trial comparing reduced intensity allogeneic hematopoietic cell transplantation to hypomethylating therapy or best supportive care in patients aged 50–75 with advanced myelodysplastic syndrome: blood and marrow transplant clinical trials network study 1102. 62nd ASH Annual Meeting, December 5–8, 2020; Abstract No. 75.
3. Giebel S, Labopin M, Houhou M, et al. Comparison of reduced intensity conditioning—allogeneic HCT and autologous HCT for elderly patients with acute lymphoblastic leukemia. An analysis from the acute leukemia working party of the European Society for Blood and Marrow Transplantation. 62nd ASH Annual Meeting, December 5–8, 2020; Abstract No. 614.
4. Zeiser R, Polverelli N, Ram R, et al. Ruxolitinib (RUX) vs best available therapy (BAT) in patients with steroid-refractory/steroid-dependent chronic graft-vs-host disease (cGVHD): primary findings from the phase 3, randomized REACH3 study. 62nd ASH Annual Meeting, December 5–8, 2020; Abstract No. 77.
5. Perrot A, Lauwers-Cances V, Cazaubiel T, et al. Early versus late autologous stem cell transplant in newly diagnosed multiple myeloma: long-term follow-up analysis of the IFM 2009 trial. 62nd ASH Annual Meeting, December 5–8, 2020; Abstract No. 143.
6. Tucci A, Re A, Pagani C, et al. Rituximab with dose-adjusted EPOCH (R-DA-EPOCH) with or without autologous stem cell transplantation (ASCT) as first line treatment in patients with aggressive B-cell lymphoma with MYC and BCL-2 and/or BCL-6 gene rearrangements or increase copy number. 62nd ASH Annual Meeting, December 5–8, 2020; Abstract No. 2131.
7. Brauer D, Backhaus D, Vucinic V, et al. Prognostic impact of nutritional status at diagnosis and weight changes in acute myeloid leukemia patients receiving allogeneic hematopoietic stem cell transplantation. 62nd ASH Annual Meeting, December 5–8, 2020; Abstract No. 2431.
8. Sallfors Holmqvist A, Chen Y, Wu J, et al. Severe/life-threatening/fatal chronic health conditions (CHcs) after allogeneic blood or marrow transplantation (BMT) in childhood—a report from the BMT survivor study (BMTSS). 62nd ASH Annual Meeting, December 5–8, 2020; Abstract No. 69.